Theoretical Computer Science, exam of 2025.01.30

SURNAME ___________________ *NAME* ____________________*PERSONAL CODE*__________

Total available time: 1h 30'. You can use books and paper notes during the exam, while electronic devices are prohibited. ***You must write the solutions on this paper alone.*** *Each correct exercise is valued 11 points.* Multichance *students do not need to answer to Exercises 2.2 and 3.2: bar here in that case* [ ]

Exercise 1

Consider the following languages defined over the alphabet {a,b}, where $\lfloor x \rfloor$ as usual represents the integer part of x.

1. $L_1 = \{a^n b^{\lfloor mn \rfloor} \mid n \geq 0, 0 < m < 5$, natural numbers}.
2. $L_2 = \{a^n b^{\lfloor mn \rfloor} \mid n \geq 0$ natural number, $m > 0$ real number}.

For each of them, construct a recognizer or a grammar with minimal power.

Exercise 2

You have been tasked with evaluating the effectiveness of generative Artificial Intelligence (AI) tools for code, such as Copilot. For this reason, you are required to create a finite and non-empty set of reference C programs and, for each of these programs, define a prompt to give to Copilot in order to generate the desired program (e.g., *"Generate a C program that computes the integer square root of an input integer"*). **Note:** A generative AI tool like Copilot returns different responses (e.g., programs) even when given the same prompt multiple times.

1) The objective is to assess whether, for each reference program, the program generated by Copilot works as intended (in the previous example, whether it correctly computes the square root of the input number) or not. Is the assigned task feasible? Provide a well-reasoned answer.

2) Suppose that, instead of studying the ability to generate C programs, you are required to study the ability to generate finite state automata. In this case, the task would involve creating a set of reference FSAs, each with its corresponding prompt, and asking the generative AI tool to produce the required automaton. Is the assigned task feasible? Provide a well-reasoned answer.

Exercise 3

Consider the language $L_1 = \{a^n b^{mn} \mid n \geq 0, 0 < m < 5$, natural numbers$\}$.
1) Describe a *k-tape Turing machine* and 2) a *RAM machine* for L, evaluating their spatial and temporal complexities, with all the available cost criteria.

space available for solutions

Solutions

Exercise 1
L_1 is constructed as $a^nb^n \cup ... \cup a^nb^{4n}$.
The recognizer with minimal power is therefore a **non-deterministic pushdown automaton** or a **context-free grammar**. For example:
$S \rightarrow A \mid B \mid C \mid D \mid \epsilon$
$A \rightarrow aAb \mid ab$
$B \rightarrow aBbb \mid abb$
$C \rightarrow aCbbb \mid abbb$
$D \rightarrow aDbbbb \mid abbbb$

L_2 is the **regular** language $a^* \cup a^+b^*$, generated by the regular grammar:

$S \rightarrow aA \mid aB \mid a \mid \varepsilon$

$A \rightarrow aA \mid aB$

$B \rightarrow bB \mid b$

Exercise 2
1) It is enough to consider the case where the set of programs contains a single element. In this case, it is necessary to be able to evaluate whether, given a program returned by Copilot in response to a prompt, it computes the same function as the corresponding program in the reference set. Clearly, the problem becomes one of equivalence between programs, which is undecidable by Rice's theorem, where the set FF of functions is the function computed by the reference program.

2) In this case, the task is feasible: it is enough to compare the language recognized by the FSA returned by Copilot with the language of each individual FSA in the reference set:

Construct the automaton A’ that recognizes the complement of the language recognized by the reference A. Then, build the automaton that recognizes the intersection of the languages recognized by A’ and the one provided by Copilot and verify if it recognizes the empty language.

Exercise 3
1. The Turing machine reads the input string, counting the a's on the first tape T1 and the b's on the second tape T2. The machine then checks if the two counters are equal, or if one is double, triple, or quadruple of the other (e.g., calculating m·T1 on a third tape and comparing with T2). Upon the first positive case, it terminates by accepting.
   The space complexity, given by the sum of the contents of the tapes, is clearly $\Theta(\log N)$, with N being the length of the input string. The time complexity requires reading the string while incrementing the counters ($\Theta(N)$), followed by checking the ratio between the counters ($\Theta(\log N)$); thus, the time complexity is $\Theta(N)$. Remember that incrementing (and decrementing) a binary counter kk times has a complexity of $\Theta(k)$, if the counter is written starting from the least significant bit, and if only the digits strictly necessary are visited with each modification of the counter.

2. The RAM machine can initially operate the reading in a manner similar to the Turing machine, with the advantage of being able to count the a's and b's using just 2 cells. After that, the machine only needs to check the ratio between the contents of the two counters, in constant time.

Complexity with the constant cost criterion:
Space: $\Theta(1)$,
Time: $\Theta(N)$.

Complexity with the logarithmic cost criterion:
Space: $\Theta(\log N)$,
Time: $\Theta(N+\log N)$ (scanning + checking ratios) = $\Theta(N)$.